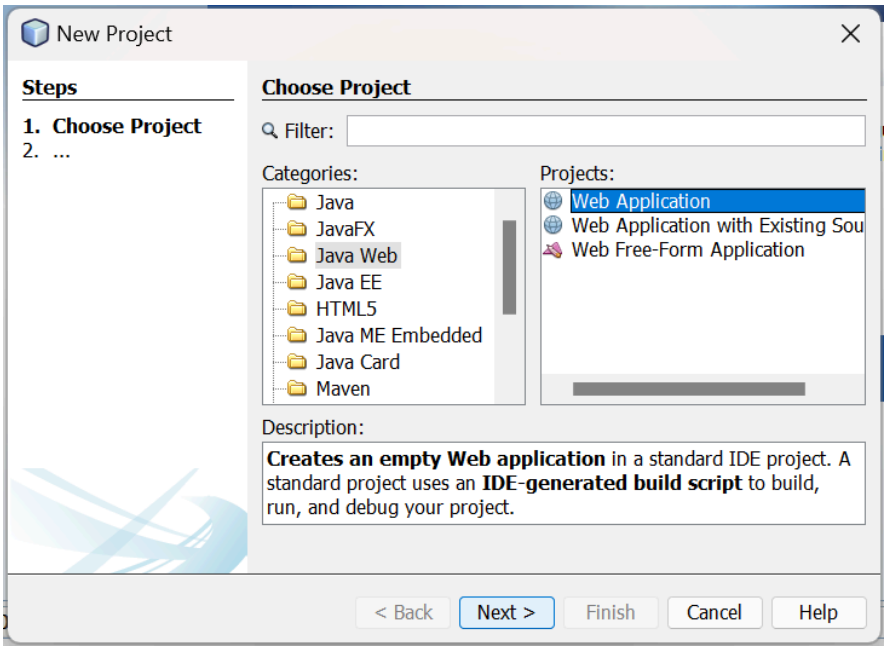
Practical No – 2

Aim: Write a program to create a simple web service that converts the temperature from Fahrenheit to Celsius and vice a versa.

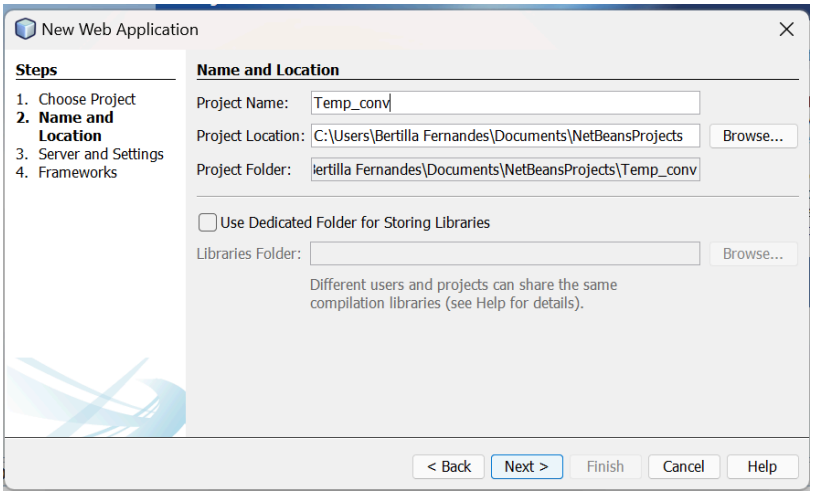
Code:

Steps:

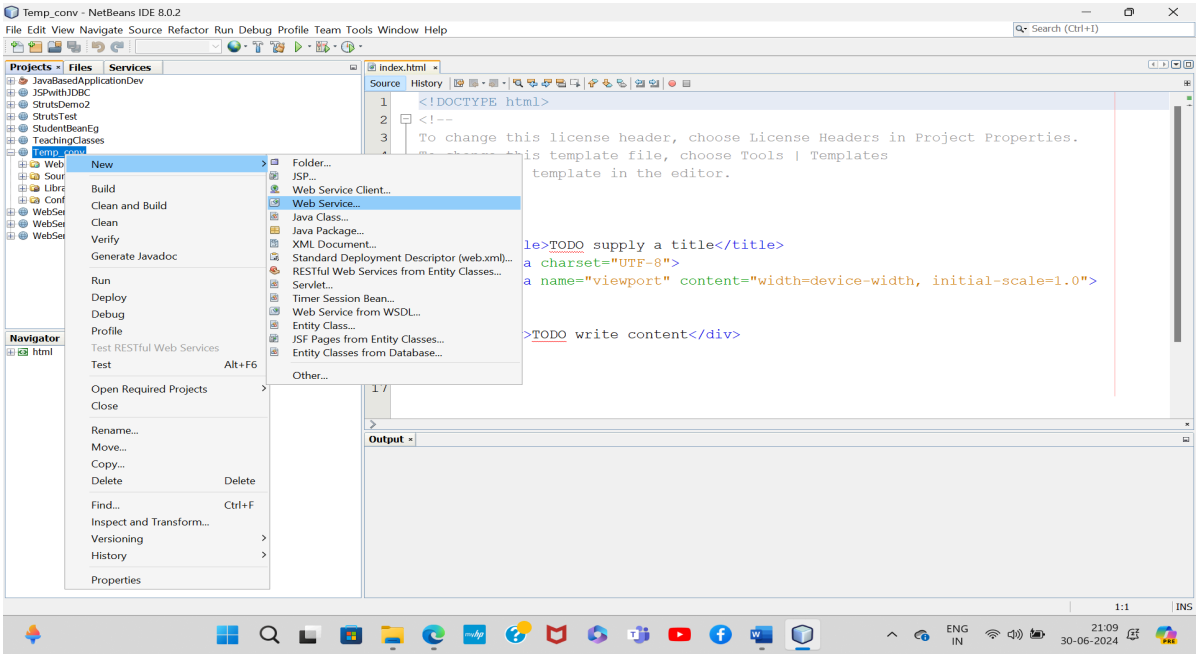
1. Go to File -> New Project. Select Java Web in categories and Web Application in Projects. Click on Next to create web based project.



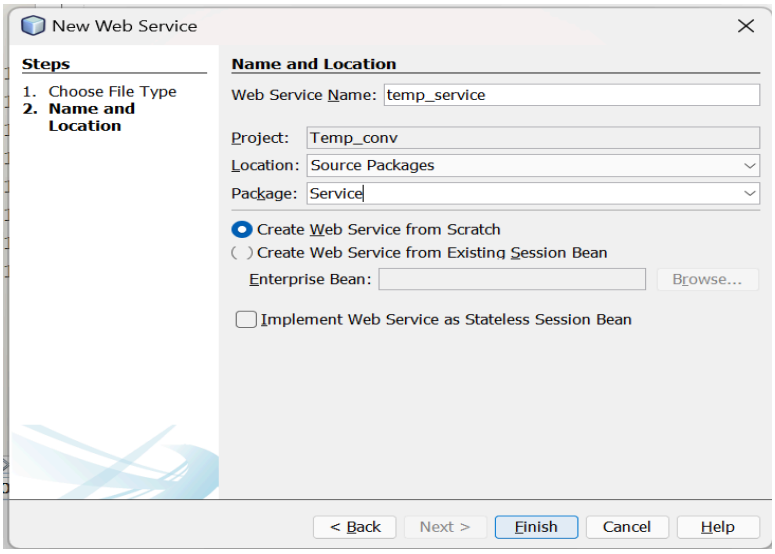
2. Enter a project name whatever you want and then click on Next. On next page click Finish.



3. Create web service. Right click on Project -> New -> Web Service



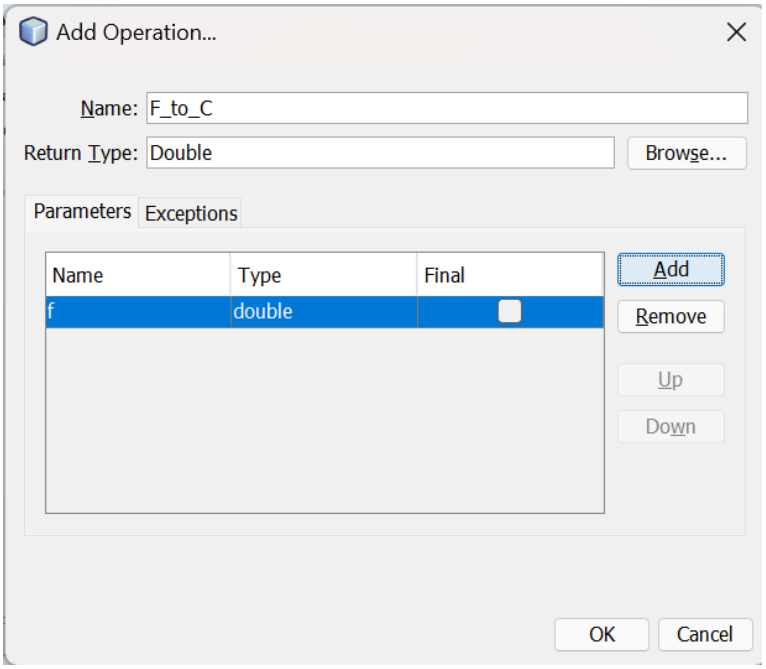
4. Enter a Web Service Name and package name and then click on Finish to create a Web Service.



5. Open temp_service.java file by double click on it, So that we can add two operation that will convert temperature from celcious to farhenheit and vice-versa. Go to design mode by click on Design.

Click on Add Operation to add operation.

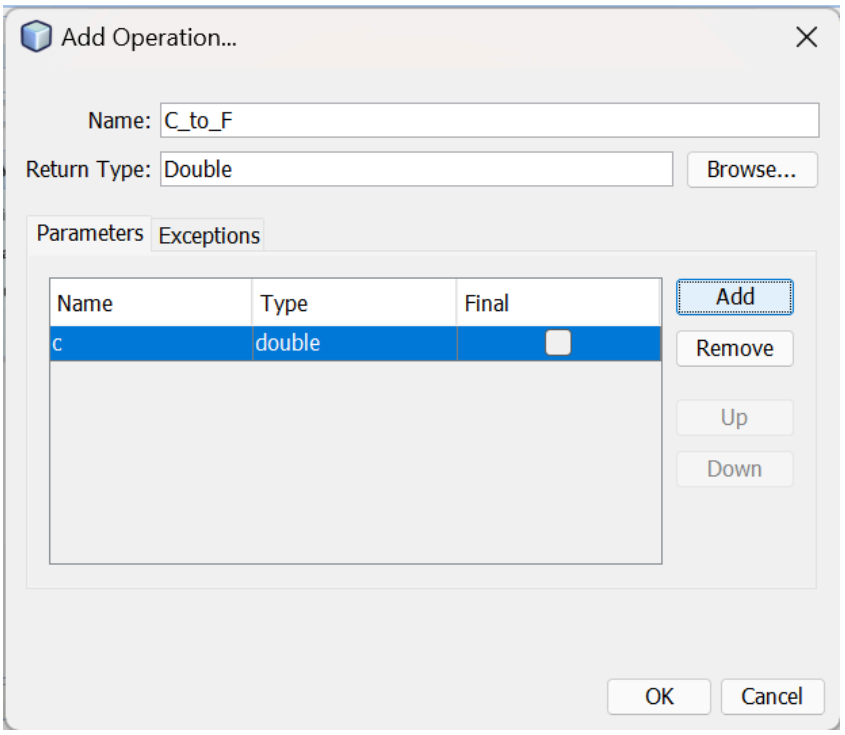
Give Operation name F_to_C and return type as Double. So this method will return value in Double data type. After that click on Add button to give parameters for method. Give its name as f and type as Double and then click on OK button.



6. Create second operation. But this time replace some above entered data with following data.

F_to_C -> C_to_F

f -> c



7. Create the following code to convert Fahrenheit to Celsius and Celsius to Fahrenheit.

Temp_service.java

package Service;

```
import javax.jws.WebService;
import javax.jws.WebMethod;
import javax.jws.WebParam;

/**
 *
 * @author Bertilla Fernandes
 */
@WebService(serviceName = "temp_service")
public class temp_service {

    /**
     * Web service operation
     */
    @WebMethod(operationName = "F_to_C")
    public Double F_to_C(@WebParam(name = "f") double f) {
        //TODO write your implementation code here:
        Double c=(f-32)*0.6;
        return c;
    }

    /**
     * Web service operation
     */
    @WebMethod(operationName = "C_to_F")
    public Double C_to_F(@WebParam(name = "c") double c) {
        //TODO write your implementation code here:
        Double f = (c*1.8)+32;
        return f;
```

```
    }
}
```

index.html × temp_service.java ×

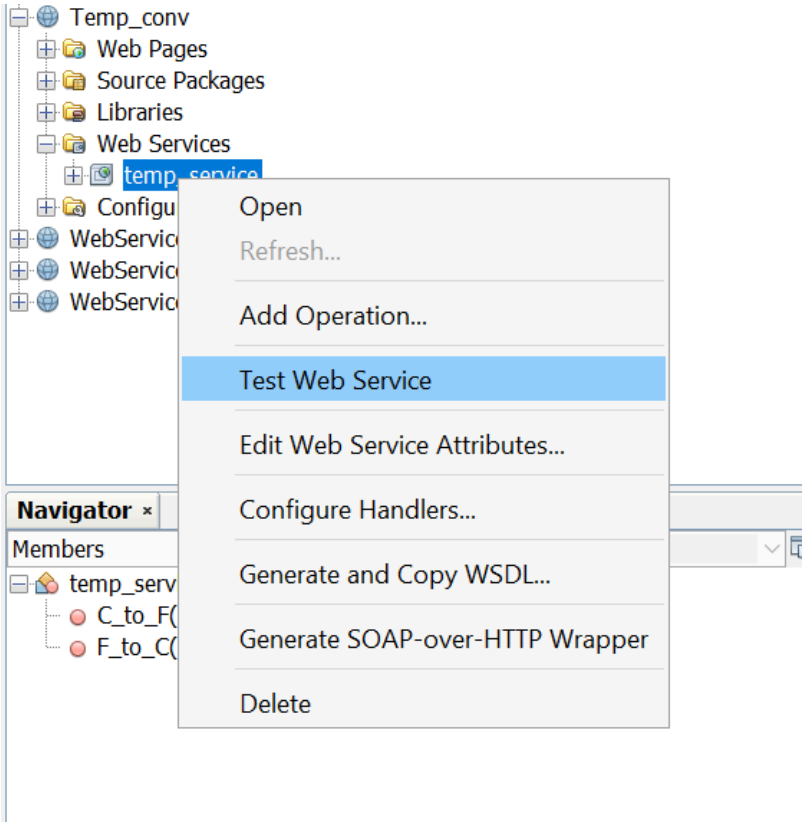
Source Design History

```
22      @WebMethod(operationName = "F_to_C")
        public Double F_to_C(@WebParam(name = "f") double f) {
24          //TODO write your implementation code here:
25          Double c=(f-32)*0.6;
26          return c;
27      }
28
29      /**
30       * Web service operation
31       */
32      @WebMethod(operationName = "C_to_F")
        public Double C_to_F(@WebParam(name = "c") double c) {
34          //TODO write your implementation code here:
35          Double f = (c*1.8)+32;
36          return f;
37      }
38  }
39
```

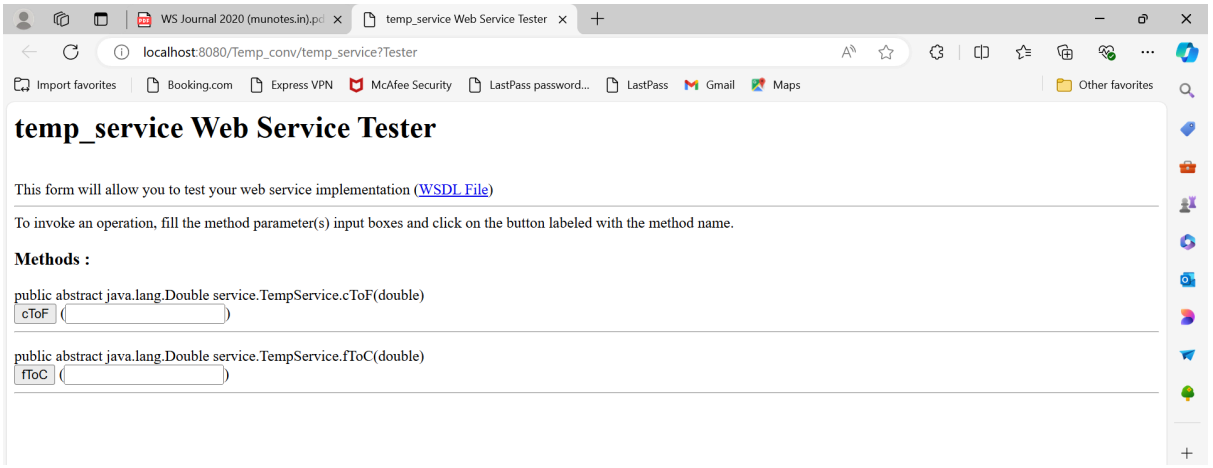
8.Now right click on project name and click on Clean and Build your project.

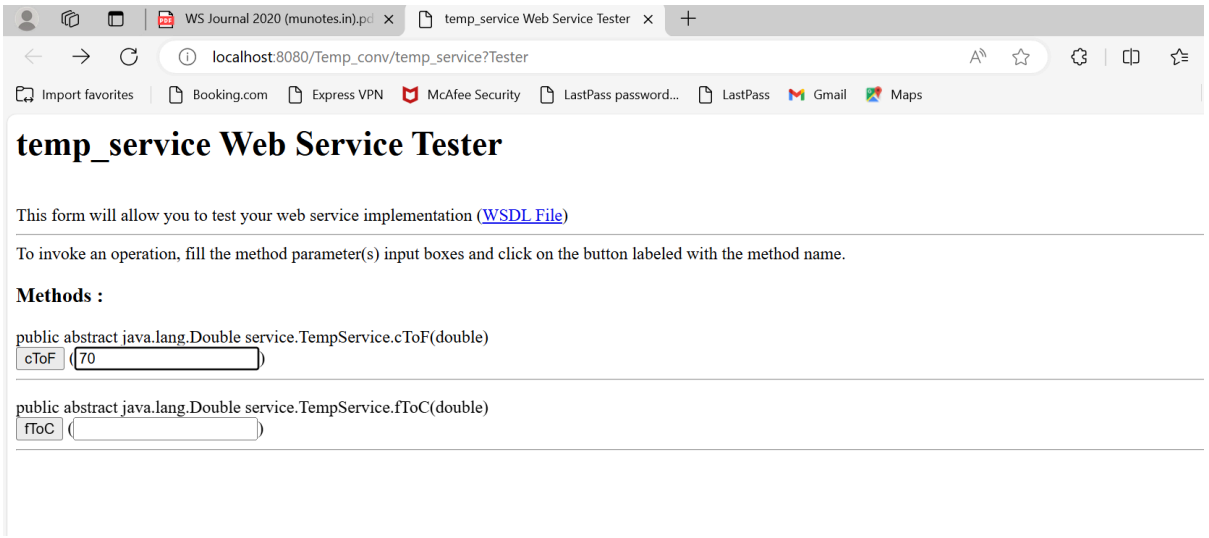
9.Next right click on project name and click on Deploy to deploy your project.

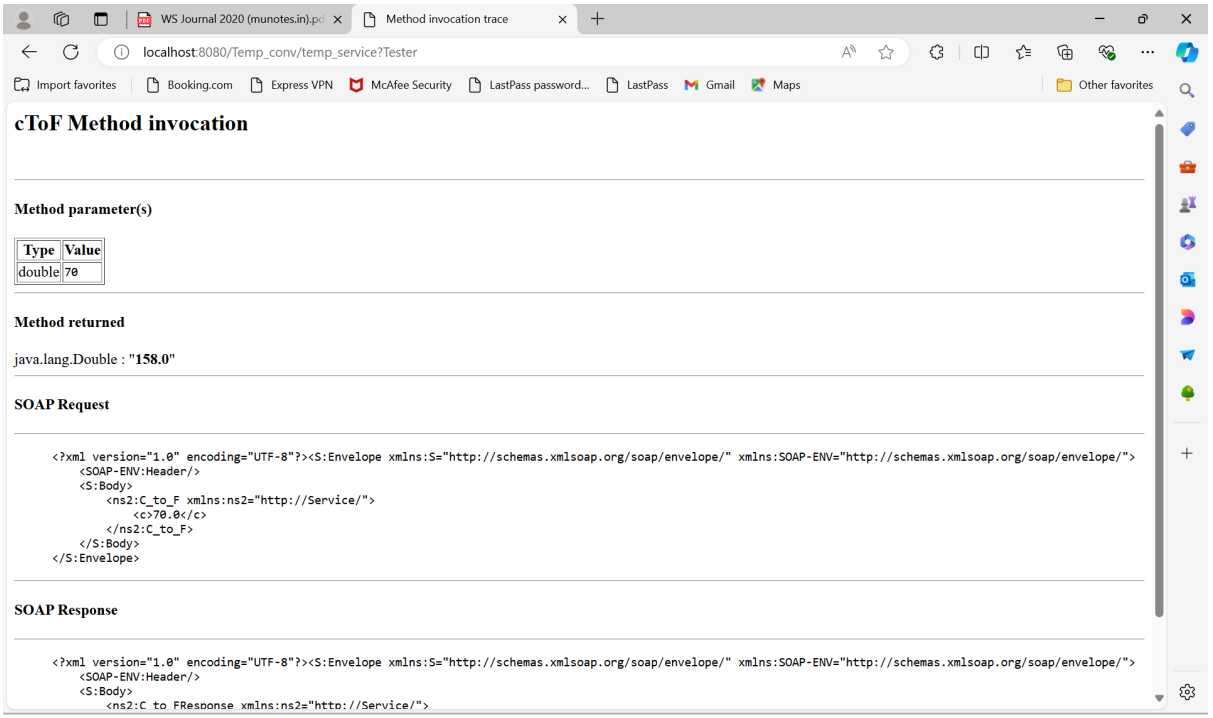
10.To test your web service follow the following process as in picture.

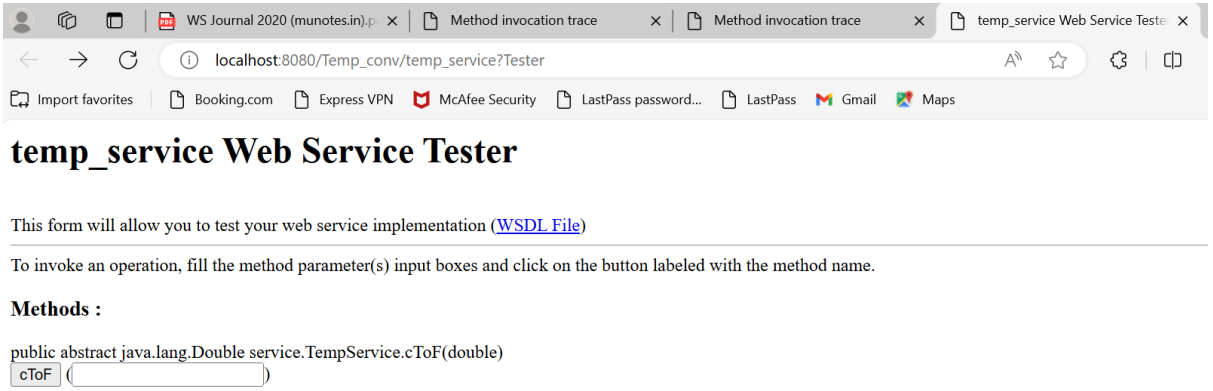


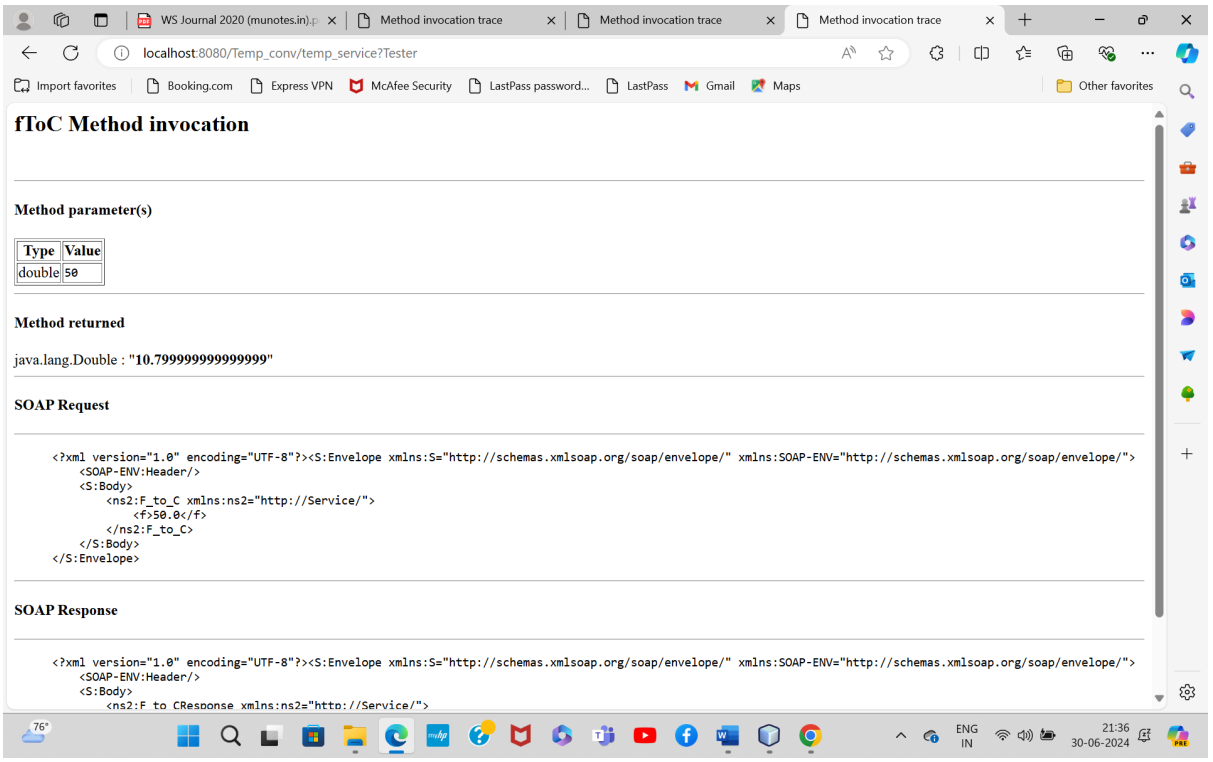
11.Following window will open in in browser. Enter the values to test the webservice.







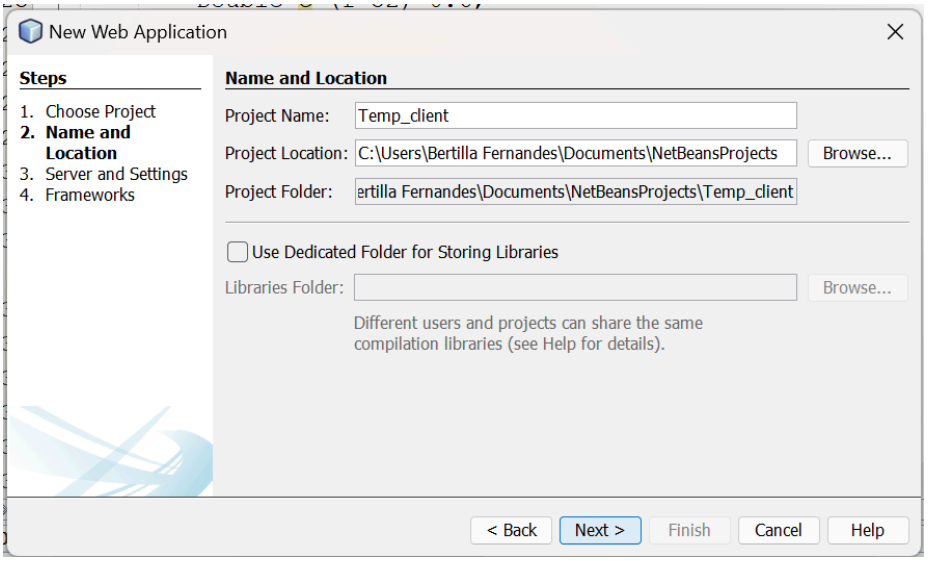




WebClient:

12. Now to consume this web service we are creating a client.

create a new web application project with name as Temp_client.



13.Now open the index.html page of Temp_client and write the following code into that.

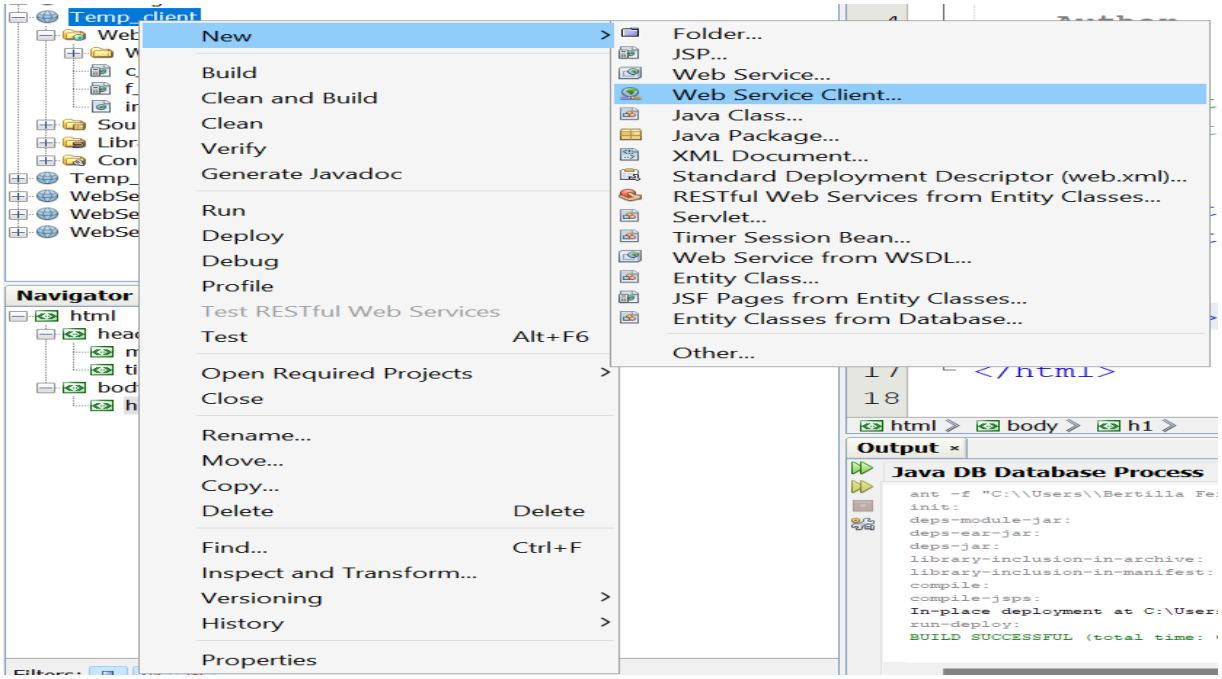
```
<html>
    <head>
        <title>TODO supply a title</title>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
    </head>
    <body>
    <form>
    <br><input type="text" name="data"><br>
    <br><input type="submit" value="Convert F to C" name="ftoc"
    formaction="f_to_c.jsp"><br>
    <br><input type="submit" value="Convert C to F" name="ctof"
    formaction="c_to_f.jsp">
    </form>
 </body>
</html>
```

14. Now create two jsp pages for both submit button. Right click on Web Pages -> New -> JSP

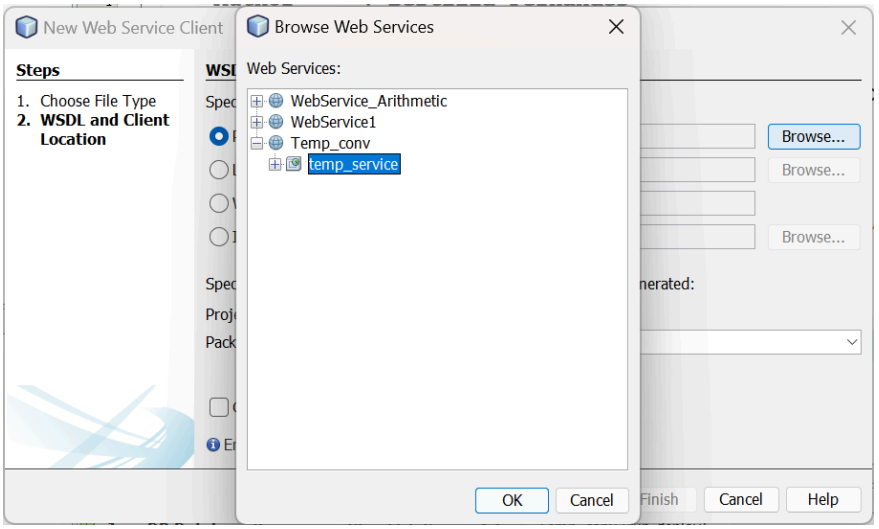
Enter file name f_to_c and then click on Finish.

Enter file name c_to_f and then click on Finish.

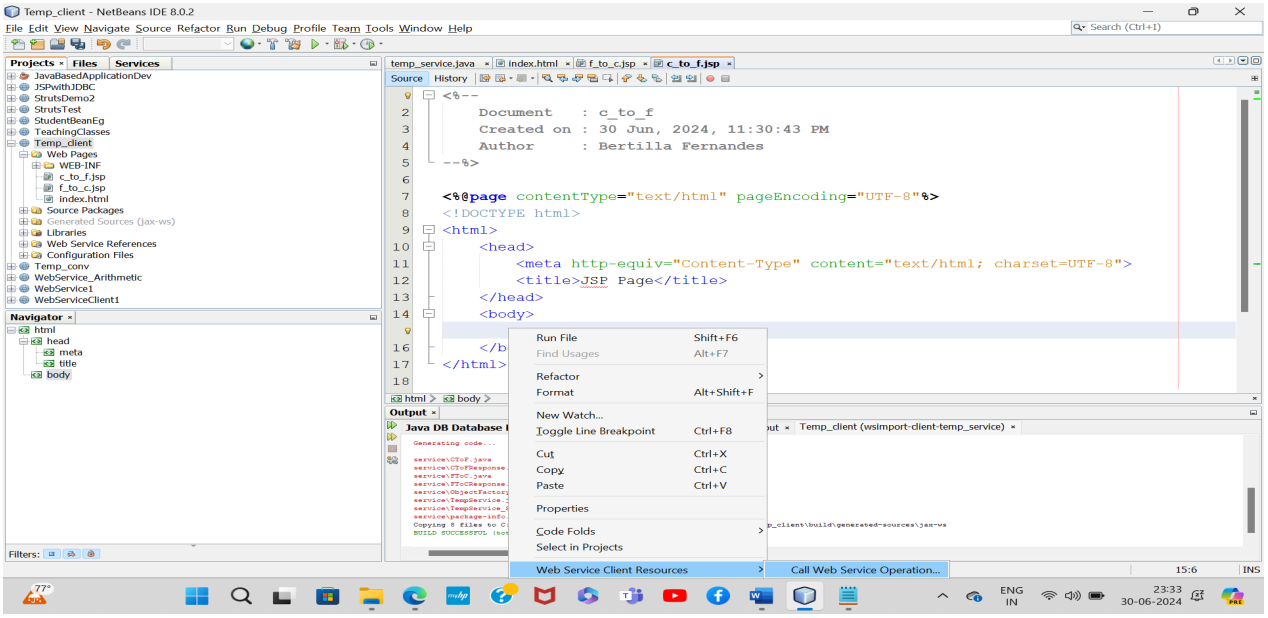
15. Right click on Temp_client and select Web Service Client as below.

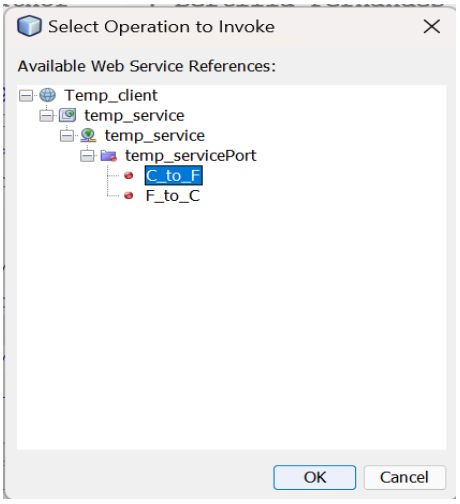


16. Click on Browse.New window will appear and then select temp_service and click on OK and then Click on Finish.

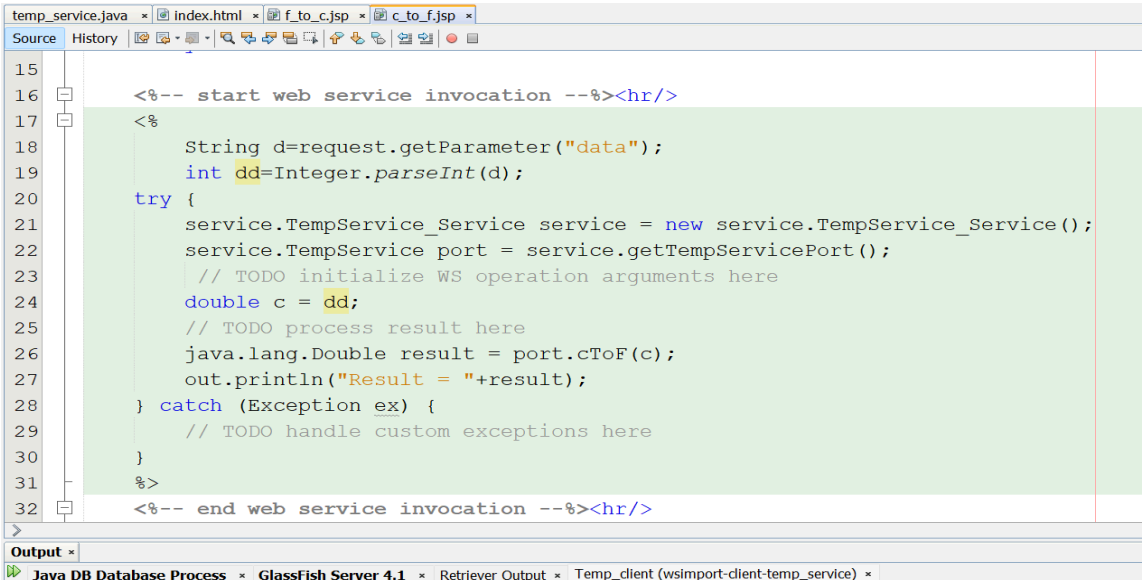


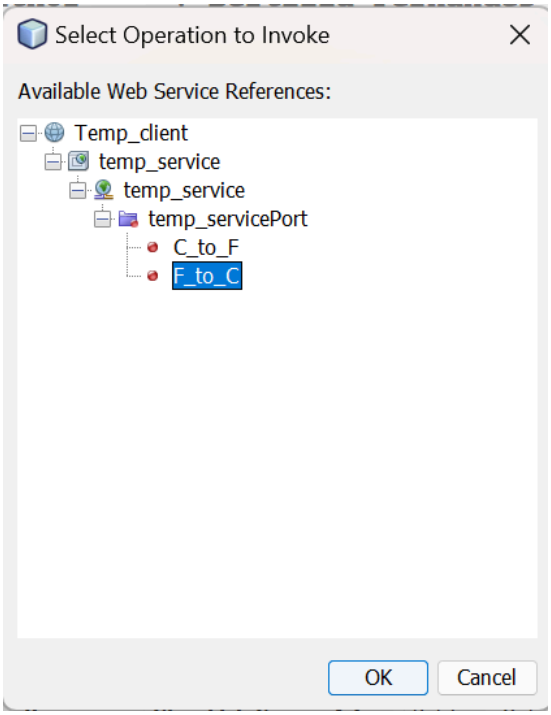
17. Now right click in between the body section and select Call Web Service Operation as below.





18. Add the code as follows to the f_to_c file.



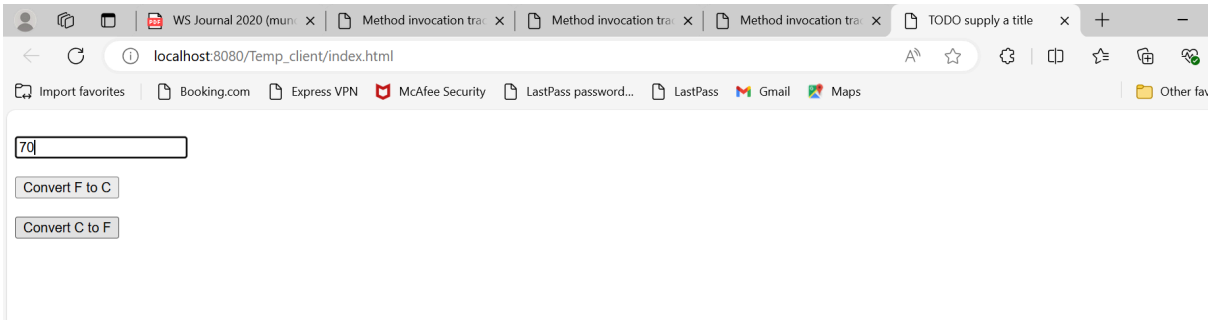


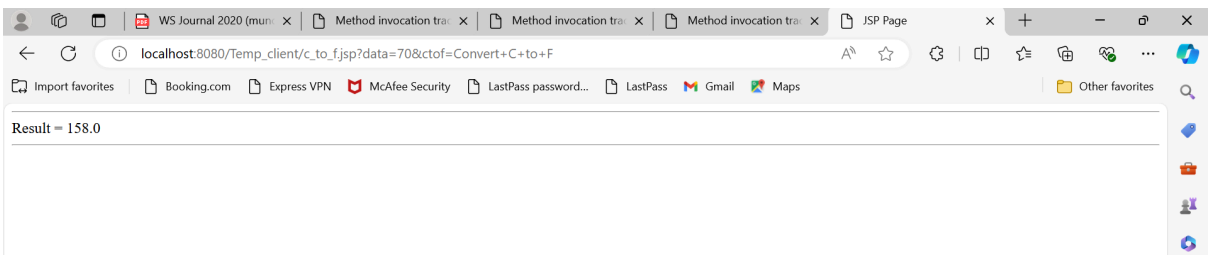
19. Add the code as follows to the f_to_c file.

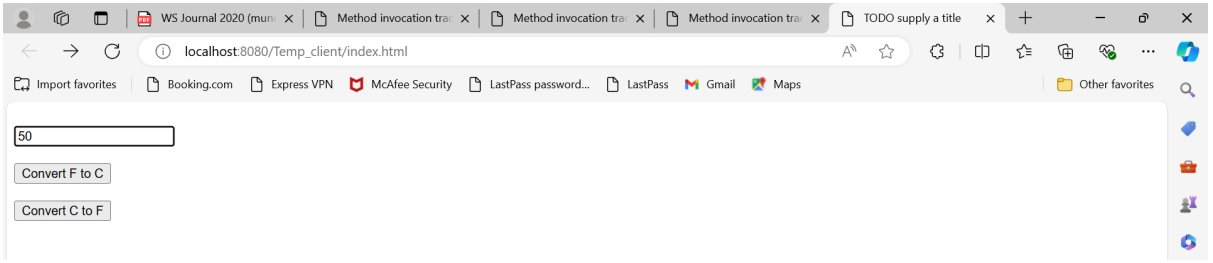
```jsp
<%-- start web service invocation --%><hr/>
<%
    String d=request.getParameter("data");
    int dd=Integer.parseInt(d);
try {
    service.TempService_Service service = new service.TempService_Service();
    service.TempService port = service.getTempServicePort();
     // TODO initialize WS operation arguments here
    double f = dd;
    // TODO process result here
    java.lang.Double result = port.fToC(f);
    out.println("Result = "+result);
} catch (Exception ex) {
    // TODO handle custom exceptions here
}
%>
<%-- end web service invocation --%><hr/>
</body>
```

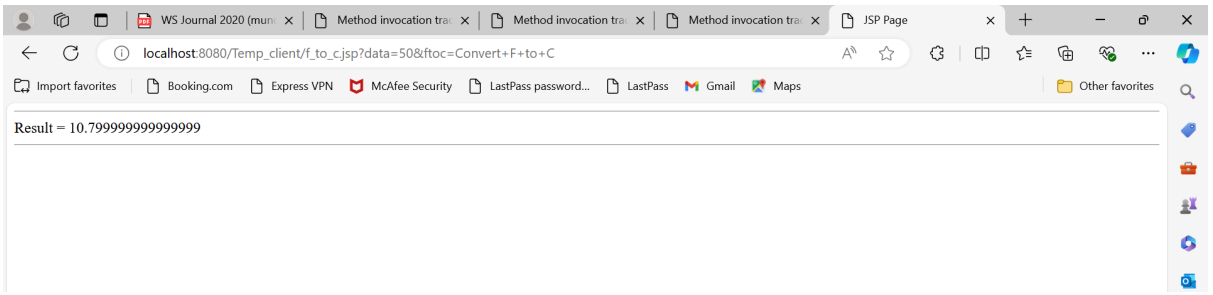
20. Now run the Temp_client project

Output:






TODO supply a title
localhost:8080/Temp_client/index.html
Import favorites
Booking.com
Express VPN
McAfee Security
LastPass password...
LastPass
Gmail
Maps
Other favorites
50
Convert F to C
Convert C to F


JSP Page
localhost:8080/Temp_client/f_to_c.jsp?data=50&ftoc=Convert+F+to+C
Import favorites
Booking.com
Express VPN
McAfee Security
LastPass password...
LastPass
Gmail
Maps
Other favorites
Result = 10.799999999999999